HR Analytics Dashboard

Where should HR act first? A Power BI project: from raw data to prioritized recommendations

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Independent portfolio project | sample dataset

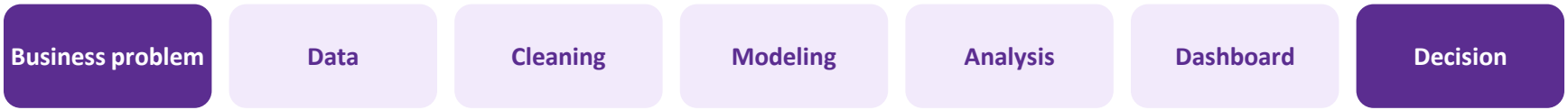


Business Problem

HR leaders need one reliable view of workforce size, hiring, satisfaction and sick leave, so they know where to act first instead of reacting to scattered reports.

The 5 questions this project answers

- How large is the workforce, and how is hiring changing?
- Which departments have the lowest satisfaction?
- Are sick days a problem in any department?
- Which hiring sources bring the most hires?
- Where should HR intervene first?



Data Cleaning (Power Query)

Clean data first, because every number on the dashboard depends on it.

What I did

- Removed duplicates and verified EmployeeID is unique
- Fixed data types (dates, numbers, text)
- Handled missing and inconsistent values
- Standardized phone numbers: removed spaces, dashes and brackets, unified the country code

Why it matters

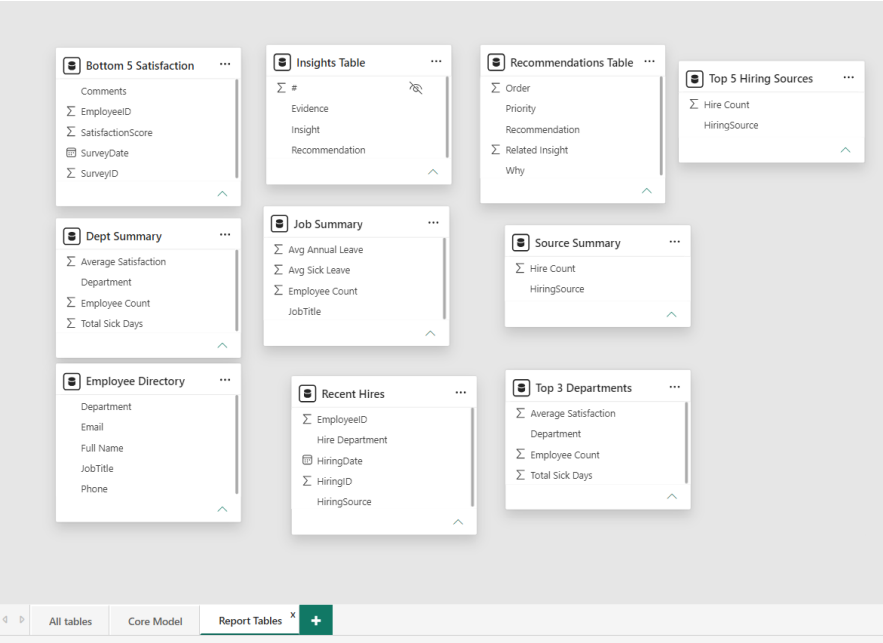
- Duplicates would inflate headcount and hires
- Wrong data types break date and number calculations
- Unique EmployeeID makes relationships between tables reliable
- Consistent formats make contact data usable

Data Model

Core Model: the tables and relationships



Report Tables: calculated with DAX, no relationships



How it works: **Employees is the central table. Hiring, Balances and Satisfaction are one record per employee (one to one). SickLeaves is one to many because an employee can have several leaves. DimDate adds year, quarter and month for time analysis.**

DAX Calculations

Measures

- Total Employees, Total Hires: counts of records
- Avg Sick Days per Employee: fair comparison across department sizes
- Satisfaction vs Company Avg: group score minus company average
- Hires YoY Growth %: change vs the previous year

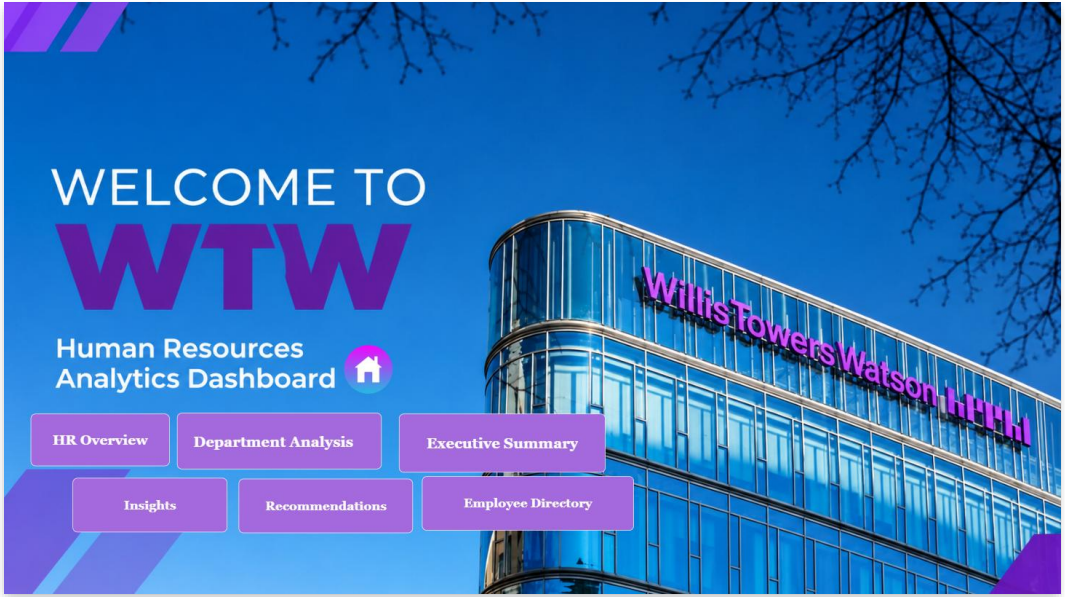
Columns and ranking

- Full Name: first and last name combined
- Hire Department, Leave JobTitle: bring department and job into other tables
- Dept Headcount Rank, Job Satisfaction Rank: RANKX ranking

Calculated tables

- Dept Summary, Job Summary
- Source Summary, Recent Hires
- Top 3 Departments, Top 5 Hiring Sources
- Bottom 5 Satisfaction
- Pre-aggregated tables feeding the report pages

Dashboard Structure



Business question: How does a manager move through the story?

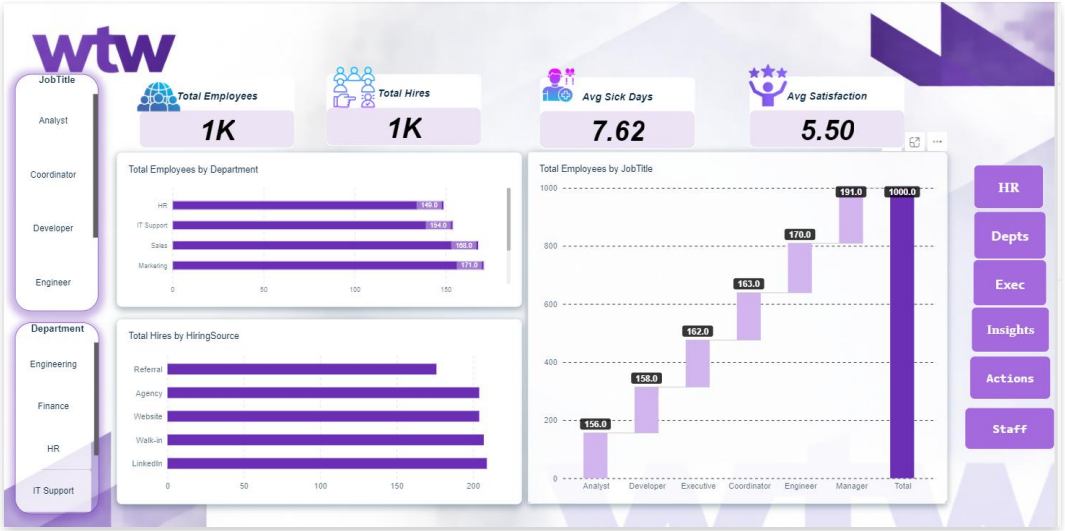
What it shows

- A Home page with navigation buttons
- One page per question: HR Overview, Department Analysis, Executive Summary, Insights, Recommendations

Key finding

The story flows from the big picture, to detail, to a decision.

HR Overview



Business question: How big is the company, and where do hires come from?

What it shows

- 1,000 employees across 6 departments
- Headcount by job title (156 to 191 each)
- Hires by source: LinkedIn, Walk-in, Website, Agency, Referral

Key finding

Referral brings the fewest hires (about 176 vs 203 to 208). This is volume, not quality, so it is not a reason to cut it.

Department Analysis



Business question: Which department needs attention?

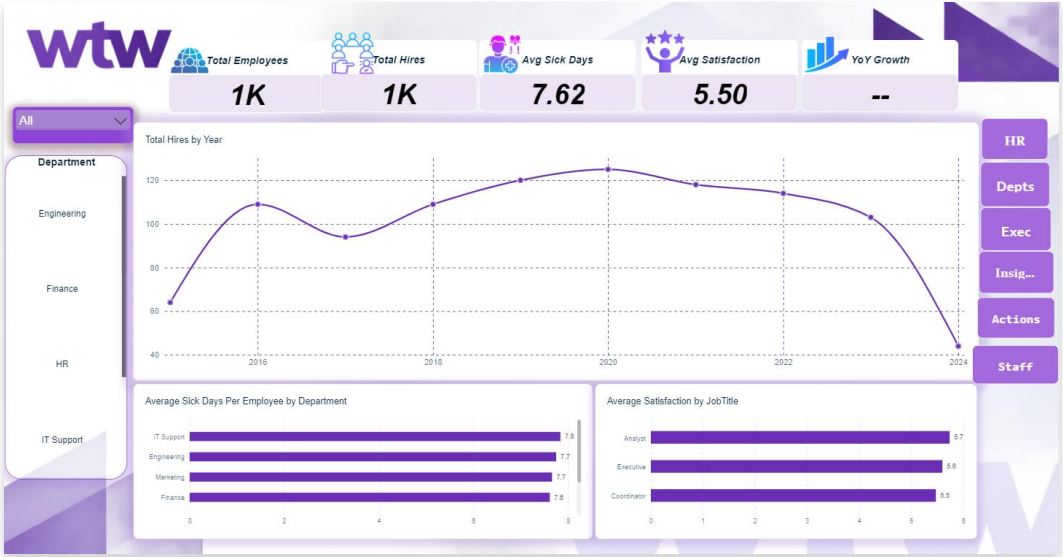
What it shows

- Satisfaction, sick days and leave balances by department
- A Satisfaction vs Company Avg gap for any selection
- Slicers by department and job title

Key finding

Engineering has the lowest satisfaction (5.06 vs 5.50) while sick days are similar everywhere (7.4 to 7.8).

Executive Summary



Business question: What is the overall picture in 10 seconds?

What it shows

- Five KPIs at the top
- Hiring trend by year
- Sick days by department and satisfaction by job title

Key finding

Hiring peaked around 2020 and then declined. 2024 may be a partial year, so the drop needs confirmation. YoY Growth appears once a single year is selected.

Key Insights

#	Insight	Evidence	So what
1	Engineering: lowest satisfaction, one of the largest teams	5.06 vs 5.50 | 179 employees	Biggest impact
2	Finance: second-lowest satisfaction	5.41 | gap about 0.09	Small gap, monitor
3	Hiring peaked around 2020, then declined	About 125 in 2020, 44 in 2024	Unconfirmed: 2024 may be partial
4	Referral is the lowest hiring source	About 176 vs 203 to 208	Volume, not quality
5	HR has the highest satisfaction	5.77 | 149 employees	Possible model, small team
6	Sick days are similar across departments	7.4 to 7.8 per employee	No action needed

Recommendations by Priority

Priority	Recommendation	Why
High	Run a survey and interviews in Engineering	Largest gap and a large headcount
Medium	Monitor Finance satisfaction	Small gap, but a large team
Medium	Review the hiring plan after confirming 2024 data	The decline may be due to partial data
Low	Evaluate candidate quality by source	Count does not equal quality
Low	Review HR practices as a possible model	Highest satisfaction, small team
Low	No department-specific sick leave policy	Differences are small

High: big impact, clear data. **Medium**: needs monitoring or data confirmation. **Low**: small impact or no action needed.

Limitations and Conclusion

Limitations

- 2024 data may be incomplete, so the hiring drop is not confirmed
- Sick day differences are small and should not be over-interpreted
- The data shows volume (hiring sources) and scores (satisfaction), not the causes behind them

Investigate Engineering first. Treat the rest as monitoring items.

Skills Demonstrated

Technical

- Power Query: cleaning and transformation
- Data modeling: star-schema-style model and relationships
- DAX: measures, calculated columns and tables, RANKX
- Power BI: KPIs, slicers, navigation, 7-page report

Analytical

- Turning business questions into metrics
- Choosing fair comparisons (averages, not totals)
- Separating facts from assumptions
- Writing prioritized, actionable recommendations

Thank you

Feedback from HR and analytics professionals is very welcome.

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Independent portfolio project using a sample dataset. Not affiliated with any company shown.